

ATTA ULLAH

Bioinformatician, ML & AI Enthusiast , Future Researcher



ABOUT ME:

Highly motivated bioinformatician, ML & AI Enthusiast with strong computational skills Python, R, ML, AI, and comprehensive knowledge in genomics, proteomics, and metabolomes and there Databases, software's, Tools Related. Also skilled in Computer-Aided Drug Design (CADD) with hand on Project. Adept at analyzing biological data, developing bioinformatics work-flows, and integrating multi-omics approaches. Exploring Machine Learning, AI and Generative AI and their applications in Bioinformatics currently at National Institutes of Health Islamabad as a junior researcher. Seeking opportunities to apply bioinformatics, computational Biology, Artificial Intelligence sectors to collaborate researchers to gain expertise in real life research opportunists to address challenges in health-care and biological research.

EDUCATION:

- **BSC (HONS) BIOINFORMATICS:**

The University of Agriculture Peshawar (<https://www.aup.edu.pk/>)

SEP / 2022 TO MAY/ 2026

Grade : 3.46 /4.00

- **INTERMEDIATE (PRE-MEDICAL):**

Gouvernent Degree College Jowar Buner

JOLY/ 2021

Grade : 75%

CONTACT INFO:

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GITHUB:

<https://github.com/Attaullah-Kahtana>

PORTFOLIO

<https://attaullah-kahtana.github.io/Atta.io/>

LOCATION:

ISLAMABAD PAKISTAN.

SKILLS AND EXPERTISE:

1: BIOINFORMATICS:

- Skilled in applying computational tools/software's to analyze biological datasets and interpret molecular patterns.
- Experienced in sequence analysis, annotation, and functional prediction workflows.
- Familiar with Biological data sets and analysis, and Omics data related tools software's.

2: PYTHON:

- Proficient in Python for data analysis, scripting, and automation of bioinformatics pipelines.
- Hands-on experience with libraries such as NumPy, Pandas, Bio python, and Matplotlib.

3: R LANGUAGE:

- Competent in R for statistical analysis and visualization of biological data.
- Familiar with packages like tidyverse, ggplot2, and Bioconductor workflows.

4: OMICS:

- Experienced in analyzing genomic sequences, variant detection, and gene annotation.
- Knowledgeable in protein sequence analysis, structure prediction, and functional annotation.
- Skilled in analyzing metabolic profiles to identify biomarkers and biochemical pathways.

5: IN-SILICO DRUG DESIGNING:

- Hands-on Experience with Structure-Based Drug Design.
- Library Preparation, Virtual Screening, Molecular Docking, and Molecular Dynamics (MD) Simulation

6: BIOLOGICAL DATA ANALYSIS:

- Experienced in handling and analyzing high-throughput biological datasets.
- Skilled in statistical analysis, visualization, and deriving meaningful biological insights.

7: NGS & NGS DATA ANALYSIS:

- Primer Designing, Real-time PCR, Sequencing, quality control, read alignment variant expression.
- and biological NGS Data Analysis interpretation.

8 : METAGENOMICS:

- NGS data preprocessing, taxonomic classification, microbial community profiling, functional annotation, diversity analysis, and biological interpretation.

9 : M L & AI:

- Familiar with applying Gen AI models for text generation, biological predictions, and automation.
- Able to integrate AI tools into research workflows for faster and smarter analysis.
- Generative AI, GenBio, Machine Learning Models, Deep learning Models. AI in Healthcare.

PROJECTS:

1: AI RESEARCH PAPER EXPLAINER (CANCERS):

HEC GENERATIVE AI HACKATHON BY PAK ANGELS:

- My team developed an innovative AI model to simplify complex cancer research papers for patients, students, caregivers, and the general public. Leveraged Generative AI and Natural Language Processing (NLP) to analyze papers, break down technical terminology, and generate clear, accurate summaries. Enabled wider accessibility of scientific knowledge while maintaining accuracy, addressing the gap between research and non-expert understanding.
- Link is here [Cancer Research Paper - a Hugging Face Space by Atta125](#)

2: STRUCTURAL BASED IN-SILICO DRUG DESIGNING FOR MULTI DRUG RESISTANT MYCOBACTERIUM TUBERCULOSIS.

- I have did my project on structural based computer based drug designing Targeting Enoyl-ACP reductase. Through molecular docking. And others computer based tools software's.
Github repository link: <https://github.com/Attaullah-Kahtana/In-Silico-Drug-Designing>.
Supervisor : Assistant Professor Dr. Muhammad Tayyab (muhammadtayyab@aup.edu.pk)

3: EARLY SEPSIS RISK PRIDECTION:

- My te a m have devolaped a machine learning model to estemate sepsis risk from ICU Patient Vital Bult from both full recourse and recourse limited hospital setting. Sepsis is a time-critical medical condition where early detection can make a meaningful difference in patient care.
Link : <https://sepsis-detection-app-4w2abxbdswgzpd4phusy7.streamlit.app/>
Github repository link: <https://github.com/Attaullah-Kahtana/Early-Sepsis-Risk-Predictor>

CERTIFICATION:

GENOMICS TECHNOLOGY:

- Training in genomics, sequencing technologies, NGS, and basic genomic data analysis, from Coursers with collaboration of the Johns_Hopkins university A u g / 2026

PYTHON:

- Python essentials/fundamentals certificates from Cisco networking academy, in collaboration with the python institute Sep / 2025

R PROGRAMMING:

- R programming for beginner's certificate from simplilearn, Aug / 2025, Sep / 2025.

NGS (Next Generation Sequencing):

- *NGS Bio-informatics for Beginners & Reproducible Research*, organized by the Biomics Platform, Institute Pasteur, under the VISION Project, October, 2025.

GENRATIVE AI:

- *HEC Generative AI Training*, conducted by Pak Angels.
- Completed hands-on training, Hackathon project ("*AI Based cancer research Paper explainer*")

ARTIFICIAL INTELLIGENCE IN BIOINFORMATICS:

- I have Completed the Artificial Intelligence in Bioinformatics course in in coursera with collaboration Taipei Medical University, gaining knowledge of how artificial intelligence and machine learning techniques are applied to bioinformatics and biological sequence analysis. Learned about data prepossessing, feature extraction and selection, machine learning and deep learning approaches, protein sequence analysis, and AI-based prediction in bioinformatics.

INTERNSHIP AT GENYX:

- During this internship, I gained valuable hands-on project experience in Bioinformatics and Computational Biology, particularly in "in silico drug design, molecular dynamics simulations, and Artificial Intelligence in healthcare and biological research. I also gained practical experience in NGS data analysis, metagenomics, Bio-python programming, and research methodology.

